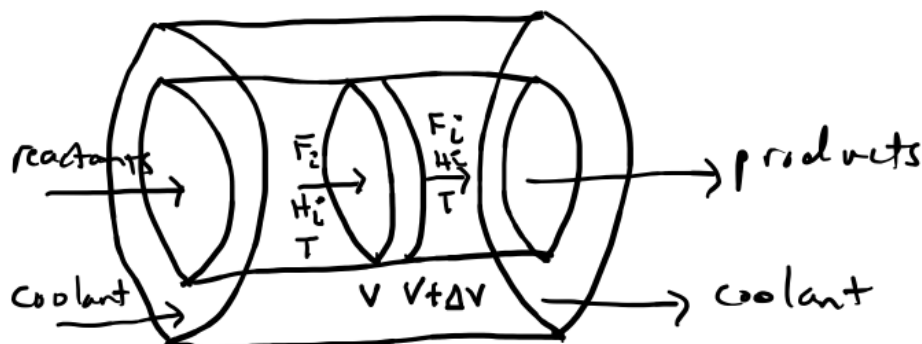


20 Energy Balances for Tubular Reactors

Let's derive the energy balance for a PFR:



Design equations now include:

- A mole balance in terms of X or one mole balance (in terms of F) for each species
- An energy balance on the reactor system
- If there is pressure drop, then the design equations also include the Ergun equation
- If the temperature of the cooling fluid is not constant, then the design equations additionally include an energy balance on the cooling fluid.

In other words, there will be between 2 and (# of species + 1) design equations if the reactor is non-isothermal.

- If there are 2 unknowns (X and T), you need 2 equations (mole balance and energy balance)
- If p is also unknown, you need the Ergun equation.
- If T_c (temperature of cooling fluid) varies, you need to write an energy balance on that system.
- If the problem cannot be solved in terms of X , you need to write mole balances for all species in the system.

Let's derive the balance for the system pictured above:

Starting from the general form of the balance: $0 = \dot{Q} + \dot{W}_s + \sum_i F_{i0} H_{i0} - \sum_i F_i H_i$:

Cancel terms: $\dot{W}_s = 0$

Applying the general form of the balance to the system:

$$0 = \dot{Q} + \sum F_{i0} H_{i0} - \sum F_i H_i$$

F_i are molar flow rates $\rightarrow$ the same F_i from the mole balance

$\dot{Q}$ is the heat transfer term, looks like $UA(T_a - T)$

Notice that the ordering inside the parentheses is intentional:

- If $T_a > T_o$ heat is transferred from shell to tube
- If $T > T_o$ heat is transferred from tube to shell

UA is overall heat transfer coefficient times area

U has units of $\frac{\text{energy}}{\text{area} \cdot \text{temp} \cdot \text{time}}$

In this class, we write $\dot{Q}$ as $Ua(T_a - T)$, where a has units of $\frac{\text{surface area}}{\text{volume}}$. This is a simplification, made so that in this class we can focus on how energy balances are constructed and how they influence reactor design rather than parameterizing heat transfer coefficients.

Then: $0 = Ua(T_a - T)\Delta V + \sum F_i H_i \Big|_V - \sum F_i H_i \Big|_{V+\Delta V}$

Apply definition of the derivative to get: $0 = Ua(T_a - T) - \frac{d}{dV} \sum F_i H_i$

Apply "product rule" to $\sum_i F_i H_i$ term: $\frac{d}{dV} \sum F_i H_i = \sum_i F_i \frac{dH_i}{dV} + \sum_i H_i \frac{dF_i}{dV}$

second
The first term is: mole balance times H $\circ$ $\sum H_i \frac{dF_i}{dV} = \sum H_i r_i$

first
The second term is: $\sum_i F_i \frac{dH_i}{dV}$

Recall that $H = H^\circ + \int C_p dT$

Then:

$$\frac{dH_i}{dV} = \frac{d}{dV} \left(H_i^0 + \int C_{p,i} dT \right) = C_{p,i} \frac{dT}{dV}$$

This development assumes C_p is constant with temperature. This is a reasonable assumption to make for liquids and gases; since we deal with liquid and gas phase reaction conditions in this class, it is a reasonable assumption to make in this class.

This leaves us with the energy balance:

$$0 = U_a(T_a - T) - \sum H_i r_i - \sum F_i C_{p,i} \frac{dT}{dV}$$

Since all r_i are related by stoichiometry, we can use Equation 3-1 (relative rates):

$$r_i = \left(\frac{\nu_i}{\nu_A} \right) (-r_A)$$

E.g., for the reaction $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$:

$$\nu_A = -1$$

$$\nu_B = -\frac{b}{a}$$

$$\nu_C = \frac{c}{a}$$

$$\nu_D = \frac{d}{a}$$

Substituting these into the energy balance:

$$0 = U_a(T_a - T) - \sum H_i \left(\frac{\nu_i}{\nu_A} \right) (-r_A) - \sum F_i C_{p,i}$$

What is $\sum_i H_i v_i$ equal to?

heat of reaction $\rightarrow \sum H_i \left(\frac{v_i}{v_A} \right) (-r_A)$ is heat of rxn per mole of A

We can prove this:

$$\Delta H^{rxn} = \sum_{\text{products}} v_i H_i - \sum_{\text{reactants}} v_i H_i$$

For the reaction $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$, $\Delta H^{rxn} = \frac{d}{a}H_D + \frac{c}{a}H_C - \frac{b}{a}H_B - H_A$.

On a per mole basis, $\Delta H^{rxn} = \sum_{\text{products}} \frac{v_i}{v_A} H_i - \sum_{\text{reactants}} \frac{v_i}{v_A} H_i$.

- This gives the heat of reaction in units of energy per mole of A.
- It is important to make sure that ΔH^{rxn} is properly normalized and that it is apparent which species it is written in terms of. Just like for r_i , it is important for accounting.

Substituting in ΔH^{rxn} :

$$0 = U_a(T_a - T) + (-\Delta H^{rxn})(-r_A) - \sum F_i C_{pi} \frac{dT}{dT}$$

Units check:

- ΔH^{rxn} is in units of $\frac{\text{energy}}{\text{mole A}}$
- r_A is in units of $\frac{\text{moles A}}{\text{volume time}}$
- The term $(-\Delta H^{rxn})(-r_A)$ is hence in units of $\frac{\text{energy}}{\text{volume time}}$

Let's re-arrange so that this formula "looks like" an ODE, this will give us the final form of the energy balance:

$$\frac{dT}{dV} = \frac{(r_A)(\Delta H^{rxn}) - U_a(T_a - T)}{\sum F_i C_{p,i}}$$

Two very important terms:

- $(-\Delta H^{rxn})(-r_A)$ is the heat generated term - due to rxn
- $U_a(T_a - T)$ is the heat removed - due to the heat exchanger

What if the reactor is adiabatic? $\dot{Q} = 0$, so we don't include $U_a(T_a - T)$ term
 $T = \frac{(-\Delta H^{rxn})X + \sum \theta_i C_{p,i} T_0 + X \Delta C_p T}{\sum \theta_i C_{p,i} + X \Delta C_p}$, $\Delta C_p = \sum |v_i| C_{p,i}^{\text{prod}} - \sum |r_i| C_{p,i}^{\text{reactant}}$
 What if T_0 is constant? supply a value - I'm done w/ my energy balance

What if T_0 is not constant?

I need to write a balance on the heat exchanger fluid

2 scenarios:

- ① Co-current coolant flow
- ② Counter-current coolant flow

① derive balance for cooling fluid

$$\frac{dT_a}{dV} = \frac{U_a(T - T_a)}{\dot{m}_c C_{p,c}}$$

$\dot{m}_c C_{p,c} \leftarrow$ coolant heat capacity
 $\uparrow$ coolant
flowrate

$$\textcircled{2} \quad \frac{dT_a}{dV} = - \frac{U_a(T - T_a)}{\dot{m}_c C_{p,c}}$$

- one more thing to account for: I don't know $T_a(V=0)$.
I know $T_a(V=V)$. So, I have an IC for T_a . I
need to guess $T_a(V=0)$ until I arrive at a value
of $T_a(V=V)$ that equals T_{a0} .